

# Miguel Beltran

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## EDUCATION

### University of California, Davis

Bachelor of Science in Computer Science

Davis, CA

Expected June 2027

**Relevant Coursework:** Algorithms & Data Structures, Operating Systems, Machine Learning, Artificial Intelligence, Computer Architecture

## EXPERIENCE

### ASUCD Picnic Day

Assistant Technical Director

Davis, CA

Nov 2025 - Present

- Led a 4-engineer Agile team to build a React/TypeScript event platform supporting 70,000+ attendees during UC Davis's annual campus-wide event.
- Designed a ranked search engine combining exact, substring, synonym expansion, and Levenshtein fuzzy matching across event schedules and locations.
- Migrated from Supabase and Google Sheets to embedded SQLite, implementing authenticated CSV synchronization pipelines with upserts and stale-record cleanup.
- Dockerized the full-stack application, reducing onboarding to one `docker compose up` command and improving local environment consistency.

### Include

Software Engineer

Davis, CA

Oct 2025 - Present

- Developed 10+ reusable Next.js components and a shared multi-page layout system for a 6-person Agile team, reducing implementation time by 80%.
- Built responsive SCSS utility mixins adopted across the codebase, standardizing mobile-first design patterns.

### Google Developer Student Club

Software Engineer

Davis, CA

Oct 2024 - Jun 2025

- Built a two-stage FAQ retrieval system using Levenshtein fuzzy matching and keyword fallback logic, reducing unanswered onboarding questions by 50%.
- Automated recurring announcements with a Python/Flask Slack bot and React/Firestore dashboard, saving organizers 2+ hours/week.

## PROJECTS

### dltracker - Deadlock Analytics Platform | Next.js 16, TypeScript, PostgreSQL, Redis

- Built a production-style analytics platform with 10 pages and 45+ reusable components consuming Steam and Deadlock APIs for analytics and leaderboards.
- Designed a 3-signal identity disambiguation system resolving up to 100 ambiguous player records per leaderboard page across 5 regions.
- Implemented Redis-backed rate limiting, Zod validation, Sentry monitoring, and multi-tier ISR caching for production-grade reliability.

### Song Popularity Predictor - ML Pipeline | Python, TensorFlow, XGBoost, scikit-learn

- Designed and evaluated a supervised ML pipeline across 32,000+ Spotify tracks using Random Forest, XGBoost, and Logistic Regression models.
- Reformulated low-signal regression tasks into binary hit classification after feature analysis.
- Engineered artist popularity and genre features with GridSearchCV tuning, improving Random Forest accuracy to 80%.

### Kitch - Kitchen Management App | Java, Spring Boot, React

- Developed a full-stack kitchen management app with a 15-endpoint Spring Boot REST API and layered MVC architecture.
- Optimized backend query performance by eliminating N+1 queries through JPQL fetch joins and transactional workflows.

### 8-Bit CPU - Custom Processor | Logisim, Computer Architecture

- Designed a 5-stage pipelined 8-bit CPU with hazard detection and data forwarding, executing 20+ custom ISA instructions.

## TECHNICAL SKILLS

**Languages:** TypeScript, JavaScript, Python, Java, C, C++, SQL, HTML/CSS

**Frontend:** React, Next.js, Tailwind CSS, SCSS, Framer Motion

**Backend:** Node.js, Express, Spring Boot, Flask, REST APIs, Prisma

**Databases & Caching:** PostgreSQL, SQLite, Redis, Firestore, Supabase

**ML & Data:** TensorFlow, PyTorch, XGBoost, scikit-learn, Pandas, NumPy

**Testing:** Playwright, Vitest

**DevOps & Infrastructure:** Docker, CI/CD, GitHub Actions, Linux, Git, Vercel, Sentry